Estimating Municipal Salt Production and Salt-Bed Counts for Pangasinan

Assumptions, equations, tests and conclusions – PangAsin capstone documentation

Prepared 24 October 2026 · Companion workbook: PangAsin_Salt_Estimates.xlsx

1. Summary

This document explains how educated-estimate (synthetic) monthly production and salt-bed counts were produced for the seven salt-producing municipalities of Pangasinan: Alaminos City, Anda, Bani, Bolinao, Dasol, Infanta and San Fabian. Every number below is reproduced by live formulas in the companion workbook.

- **Annual baseline (2025): 9,820.2 MT.** Alaminos City (1,099 MT) uses its primary master-sheet production (column U read as kg per month × 6 months), which agrees with its FOI value (1,070 MT) within 3% (Test T10). San Fabian uses its primary record (106 MT, with FOI filling 2026), and the other five, including Bani, BFAR FOI figures. Bani's master-sheet reports (8,160 MT) are a switch.
- **Bed yield x = 144.36 kg per salt bed per year, not 310.61.** The handwritten calculation divides by 3,444 beds, which is the running total of only the first 21 Alaminos rows. The full master sheet holds 7,410 beds. This halves x and doubles every estimated bed count (Test T2). A primary-based calibration (1,099,260 ÷ 7,410 = 148.35 kg/bed) agrees with the FOI-based value within 3% (T10).
- **Estimated salt beds (2025): about 67,819 in total.** Alaminos City (7,410) is observed; Bolinao ≈ 30,907, Infanta ≈ 9,951, Anda ≈ 9,228, Dasol ≈ 5,378, Bani ≈ 4,210 and San Fabian ≈ 734 bed-equivalents are estimated.
- **Monthly method: the observed seasonal profile (equation 5a) for every municipality, so no month is empty.** Recorded seasons (San Fabian Nov–May, Alaminos Dec–May, Bani Jan–Apr) are switches. The linear-arithmetic idea (M1 = 8%) was built and tested. It misallocates 53% of annual output across months, no better than a flat 1/12 split, so it is kept only as a selectable comparison (Test T7).
- **Main limitation:** FOI values are lower bounds (the FOI provincial total is about 15% of the 2021 provincial-government total), and one yield constant from Alaminos cannot be assumed to fit larger Dasol-type beds. Estimates carry confidence tiers for this reason.

2. Data inventory and source hierarchy

Dataset	Type	Content	Used for
salt_MasterSheet1 – Alaminos City	Primary	38 survey rows: salt beds, bed size, bed area, production rate (col. U)	Bed count, bed area, calibration of x; 2025 output (col. U read as per month × 6)
salt_MasterSheet1 – Bani	Primary	5 farms: area, 2024–25 output (MT), production period Jan–Apr	Alternative annual source and Jan–Apr window (selector)
San Fabian email screenshots	Primary	2023–2025 output, production area, season Nov–May, cooked salt	Annual production; optional Nov–May window
Handwritten cards (Linear-Arithmetic PDF)	Working notes	x, bed counts, San Fabian consistency, M1 = 8%	Method definitions; reconciled in Appendix A
FOI RS3969 (BFAR RFO1)	Secondary	Active producers; provincial monthly volume 2023–2025	Seasonal profile
FOI RS4760 (BFAR RFO1)	Secondary	Municipal production 2024–2026 (assisted + non-assisted)	Annual baseline where no primary data

Dataset	Type	Content	Used for
OPAg 2021 (Pangasinan Provincial Gov.)	External benchmark	Municipal production 2021, farm area, banigan count	Tests T3 and T9 only – never feeds an estimate

Rule R1 (source hierarchy). Primary data override secondary data; secondary data are used only where no primary value exists; external benchmarks are used to test, never to estimate. **One exception by decision:** Bani is treated like the other FOI municipalities (FOI annual values, observed seasonal profile); its master-sheet reports remain available through Assumptions!B11. Alaminos 2025 output follows its primary master-sheet rate, with FOI as a switch (Assumptions!B23; see T10).

3. Thesis and assumptions

Thesis. The most reliable municipal estimates obtainable from the present data come from (i) letting primary records override secondary ones, (ii) calibrating one bed-yield constant only where beds were actually counted, (iii) allocating annual totals to months using only observed seasonal evidence, and (iv) reporting the uncertainty that remains. Each rule is kept only if it survives a numeric test; "most reliable" therefore means best supported by the available evidence, not proven accurate.

ID	Assumption	Why it is the best available choice	Tested by
A1	Primary values override FOI where both exist (Bani excepted by decision)	Primary records come from the producers; FOI is a partial secondary tally	T1, T4
A2	Where only FOI exists, the FOI value is the baseline (a lower bound)	No better annual figure is in the dataset; no coverage correction is supportable from two comparisons	T1, T9
A3	x = FOI 2025 ÷ surveyed beds, Alaminos City only	Only municipality with a bed census; 2025 is the only year whose FOI note says assisted + non-assisted	T2, T3, T6
A4	x transfers to the other municipalities	Needed to estimate beds at all; similar solar-bed size assumed	T5, T9
A5	Months follow the observed provincial profile of the same year for every municipality; recorded seasons are optional windows	The only monthly series is provincial; primary notes give exact seasons	T7, T8
A6	2026 FOI data are provisional	Cut-off date unknown; Bani has no primary 2026 value	Used only for sensitivity

4. Equations and solutions

4.1 Annual production

$$Y_{m,t} = c_m \cdot S_{m,t} \qquad (1)$$

where S is the primary value P if one exists for municipality m and year t, otherwise the FOI value F; c is an optional coverage multiplier (default 1.00). Solution for 2025: Alaminos S = 183,210 kg per month (master column U) × 6 months = 1,099,260 kg (FOI: 1,069,740 kg); San Fabian S = 106,000 kg (primary); the other five, including Bani (607,800 kg), take F. Total = 9,820,225 kg. By decision Bani is treated like the other FOI municipalities; its master-sheet value (8,160,000 kg) is available through Assumptions!B11 = 2.

San Fabian hybrid series. The primary record has no 2026 value ("ongoing data collection") and the FOI file has no 2023 value, so San Fabian is built as a hybrid: primary where it exists, FOI where it is missing (Table 1a). The 2024 and 2025 values come entirely from primary data. The 2026 gap is filled with the FOI value as reported (91,975 kg, a provisional partial-year figure). Assumptions!B24 = 2 instead scales that FOI value by the primary ÷ FOI ratio of the two overlapping years, 209,500 ÷ 166,495 = 1.258, giving 115,732 kg. Monthly

values follow equation 5a: the observed profile for 2024–2025 and the pooled 2024–25 profile for 2023 and 2026, which have no full provincial monthly series (Table 4a; workbook Monthly_Estimates rows 30–35).

Table 1a. San Fabian hybrid annual series (kg)

Year	Primary	FOI	Hybrid used	Basis
2023	84,000	–	84,000	Primary only (no FOI)
2024	103,500	96,000	103,500	Primary
2025	106,000	70,495	106,000	Primary
2026	–	91,975	91,975	FOI fill (primary ongoing)

4.2 Bed-yield calibration

$$x = F_{ALA,2025} \div B_{ALA} = 1{,}069{,}740 \text{ kg} \div 7{,}410 \text{ beds} = 144.36 \text{ kg/bed/yr} \qquad (2)$$

F is the FOI Alaminos City 2025 volume and B the surveyed bed count (Σ of 38 rows, Alaminos_Beds sheet). Dividing by the legacy count of 3,444 gives 310.61. Two further calibrations bound the sensitivity: x_{2026} = 1,327,670 ÷ 7,410 = 179.17 (provisional) and their mean 161.77. The workbook default is x_{2025} because 2026 is incomplete. A primary-based value, 183,210 kg a month × 6 ÷ 7,410 = 148.35, is selectable (Assumptions!B4 = 4).

4.3 Salt-bed estimate

$$\hat{N}_m = Y_{m,2025} \div x \qquad (3)$$

Worked example, Anda: $\hat{N}$ = 1,332,211 ÷ 144.36 = 9,228 beds (the legacy basis gives 1,332,211 ÷ 310.61 = 4,289.02, matching the handwritten card). Alaminos City is the observed count, not an estimate. The uncertainty band divides by the higher and lower of x_{2025} and x_{2026}.

4.4 Monthly allocation

$$y_{m,t,j} = Y_{m,t} \cdot s_{m,t,j} \qquad (4)$$

The share s comes from one of three definitions:

$$s_{t,j} = Q_{t,j} \div (Q_{t,1} + Q_{t,2} + \ldots + Q_{t,12}) \qquad (5a)$$

Observed profile: Q is the FOI provincial monthly volume (2024 and 2025). Applied to every municipality by default.

$$s_j = f_j \div (f_1 + f_2 + \ldots + f_{12}) \qquad (5b)$$

Window profile: f = 1 for months inside the primary-stated production period, 0 otherwise. San Fabian = Nov–May (14.29% per month), Alaminos = Dec–May (16.67%) and Bani = Jan–Apr (25%). These windows are workbook switches and are not applied by default, because they leave the other months empty.

$$s_j = a_1 + (\mathrm{j} - 1)\cdot \mathrm{d}, \quad \mathrm{d} = (1 - 12\cdot a_1) \div 66 \qquad (5c)$$

Linear arithmetic: the twelve shares must sum to 1, so $12a_1 + 66d = 1$. With the card's a_1 = 8%, d = 0.04 ÷ 66 = 0.0606 percentage points, giving 8.00% in January rising to 8.67% in December. Anda January = 0.08 × 1,332,211 = 106,577 kg (card: 106,568).

Table 1. Monthly shares of annual production (%)

Profile	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Observed 2024	9.4	1.8	16.7	24.6	29.8	10.9	1.2	1.2	1.2	1.1	1.1	1.1
Observed 2025	0.2	0.6	5.3	37.3	28.5	14.2	4.0	0.9	1.6	1.9	2.3	3.0
Pooled mean	4.8	1.2	11.0	31.0	29.1	12.5	2.6	1.1	1.4	1.5	1.7	2.0
Linear (a_1=8%)	8.0	8.1	8.1	8.2	8.2	8.3	8.4	8.4	8.5	8.5	8.6	8.7

Observed 2024 comes from a file noted "assisted associations only"; 2025 from the file covering assisted and non-assisted associations. 2023 has July–December only and is not used.

5. Testing

Ten tests were run on the primary and secondary data. All are live formulas on the Testing sheet. Verdicts: Supports (the assumption survives), Flag (an issue was found and handled), Rejects (the alternative fails).

Table 2. Test summary

Test	Question	Result	Verdict
T1	Do FOI and primary data agree?	FOI is 93% and 67% of primary for San Fabian 2024 and 2025, and 7% for Bani	Supports A1, A2
T2	Is the handwritten denominator the full census?	3,444 = running total through row 21 of 38; full sheet = 7,410 (×2.15)	Flag – corrected
T3	Is x physically plausible?	40.7, 40.0 and 44.8 t/ha from three independent sources, within 12%	Supports A3
T4	Which source is steadier across years (San Fabian)?	CV falls from 15.9% (FOI only) to 7.4% with primary data	Supports A1
T5	Do implied beds fit reported area?	Bani 7% of area on FOI basis, 98% on primary (consistent); San Fabian not applicable (cooked salt)	Flag – bed-equivalent
T6	Did beds really grow 2024 → 2025?	Bolinao ×204, Anda ×11.5, province ×4.3: reporting expansion	Supports use of 2025
T7	Which monthly method fits observed shares?	Misallocated output: linear 52.9%, uniform 52.3%, empirical 23.3%	Rejects linear as default
T8	Do methods respect primary production windows?	Linear puts 67.6% of Bani output outside Jan–Apr (primary basis); windows 0%	Supports A5
T9	Does FOI cover the real industry?	FOI province 2025 ≈ 15% of OPAg 2021; Dasol ≈ 3%; Dasol banigan ≈ 17× an Alaminos bed	Flag – lower bound
T10	Are Alaminos master-sheet rates credible as production?	Col. U as monthly: 1,099 MT, +2.8% vs FOI, 41.8 t/ha; col. V reading 30.8× FOI	Supports (unit to confirm)

T1 Source concordance

Municipality	Year	Primary (kg)	FOI (kg)	FOI ÷ primary	FOI shortfall
San Fabian	2024	103,500	96,000	92.7%	7.3%
San Fabian	2025	106,000	70,495	66.5%	33.5%
Bani	2024	9,444,690	649,400	6.9%	93.1%
Bani	2025	8,160,000	607,800	7.4%	92.6%

FOI is below primary in every comparison, by 7% to 93%. This supports rule R1 and warns that FOI-only municipalities are probably under-stated.

T2 Denominator audit

Cumulative Alaminos beds reach exactly 3,444 at row 21 (Jaime Capitle). Seventeen later rows (Jaime Quezada through the two unsurveyed groups) add 3,966 beds, so the full count is 7,410 and the full-sheet yield is 144.36 kg/bed/yr. The master sheet appears to have been extended after the handwritten calculation. The workbook defaults to the full sheet and keeps the legacy basis as selector option 3.

T3 Physical plausibility of x

Independent yield check	Value (t/ha/yr)	Source
Alaminos 2025: FOI ÷ surveyed bed area (26.29 ha)	40.7	FOI RS4760; master sheet
Bani 2025: reported output ÷ active area (204 ha)	40.0	Master sheet

Independent yield check	Value (t/ha/yr)	Source
Pangasinan 2021: 64,156 MT ÷ 1,432.4 ha	44.8	OPAg via pangasinan.gov.ph

The three yields agree within 12%, and x = 144.36 kg per 35.5 m² bed equals 4.07 kg/m²/yr (40.7 t/ha). By contrast, the master sheet's per-day columns (T–X) sum to 5,496,300 kg for a single month, 5.1 times the whole FOI year and about 209 t/ha in one month. Column V (U × 30) is therefore not credible. Column U, labelled per day, is plausible read as kg per month, which is how it is used (T10). Note that Bani's 2025 output equals exactly 40 t/ha × area for every active farm, so that figure looks like an area norm rather than a measured harvest.

T4 Temporal stability (San Fabian)

Series	Mean (kg)	Std dev (kg)	CV	Note
A · FOI only (2024–2026)	86,157	13,712	15.9%	
B · Card: FOI 2024, primary 2025, FOI 2026	97,992	7,222	7.4%	card
C · Strict hierarchy: primary 2024 and 2025, FOI 2026	100,492	7,481	7.4%	used
D · Primary only 2023–2025	97,833	12,045	12.3%	growth

Replacing FOI with primary values halves the variation. Because $\hat{N} = Y \div x$, the CV of the bed estimate equals the CV of production, so this result does not depend on x. San Fabian's area rose from 0.04 to 0.06 ha in 2025, which can legitimately lift output (series D).

T5 Area consistency of implied beds

Implied bed area = $\hat{N} \times 35.48$ m². Alaminos is 26.29 ha by construction. Bani implies 14.9 ha against 204 ha reported (7%) on the FOI basis, which is consistent but uninformative; on the primary basis it implies 200.6 ha (98%), consistent only if Bani's "total area" is net bed surface. San Fabian implies 2.6 ha against 0.06 ha reported (43×). Cooked (brine-boiling) salt is not area-bound, so San Fabian's 734 is a production-equivalent number of solar-size beds, not a physical count. A literal area-based count would be about 17.

T6 Year-over-year coverage and the 2024 → 2025 question

FOI province totals rise ×4.3 from 2024 to 2025 (2.27 to 9.76 million kg). Bolinao rises ×204 and Anda ×11.5, which no bed construction can explain. Applying x to Alaminos 2024 output implies 3,092 beds against 7,410 surveyed. The data cannot prove whether Alaminos built beds, but the cross-municipality pattern points to wider reporting. Hence 2025 is used for calibration and 2024 bed counts are shown only as flagged lower bounds.

T7 Monthly method comparison (leave-one-year-out)

Each method predicts the monthly shares of a year; the error is measured against that year's observed provincial shares. "Misallocated" is half the sum of absolute share errors, i.e. the fraction of annual output placed in the wrong month.

Method	Year	Mean abs. error (pp)	Misallocated	Mean
Uniform 1/12	2024 / 2025	8.28 / 9.16	49.7% / 55.0%	52.3%
Linear arithmetic (a_1 = 8%)	2024 / 2025	8.41 / 9.21	50.5% / 55.3%	52.9%
Empirical profile from the other year	2024 / 2025	3.88 / 3.88	23.3% / 23.3%	23.3%

The linear method is slightly worse than a flat split because production is concentrated: April alone holds 25–37% of annual output in the observed years, while the linear method gives 8.2%. January is 0.24% (2025) or

9.4% (2024), so the 8% anchor matches only one of two years. An empirical profile, even from the wrong year, halves the error. The workbook keeps linear arithmetic as selector option 3 for comparison.

T8 Primary production windows

Method	Bani outside Jan–Apr	San Fabian outside Nov–May
Linear arithmetic	67.6%	42.1%
Uniform 1/12	66.7%	41.7%
Observed provincial 2025 profile	56.5%	22.7%
Primary window (flat inside window)	0%	0%

Even the observed provincial profile places over half of Bani's output outside its recorded Jan–Apr season, so a primary window is applied to San Fabian (and to Bani if its primary basis is selected). Within a window the split is flat because no monthly evidence supports any other shape (maximum-entropy choice).

T9 External benchmark

Municipality	OPAg 2021 (MT)	FOI 2025 (MT)	FOI ÷ OPAg	Comment
Alaminos City	5,208	1,069.7	20.5%	
Anda	3,750	1,332.2	35.5%	
Bani	4,617	607.8	13.2%	primary: 8,160
Bolinao	12,240	4,461.9	36.5%	
Dasol	24,000	776.4	3.2%	10,000+ banigan
Infanta	9,622.8	1,436.6	14.9%	not from beds
Province	64,156	9,755.2	15.2%	

The years differ (2021 versus 2025), so the ratios are indicative, but FOI is clearly a partial tally. The OPAg Dasol record also gives 24,000 MT from more than 10,000 banigan, 2,400 kg per unit or 16.6 times the Alaminos x. Dasol units are therefore much larger than 35 m² Alaminos beds, and our Dasol figure (5,378) is a count of Alaminos-size bed-equivalents. Its closeness to 10,000 is partly coincidence: FOI under-coverage and bed-size difference push in opposite directions.

T10 Alaminos master-sheet production rate

The Alaminos master sheet leaves annual production empty (the 2024 and 2025 columns have no values). Its only primary production figures are per-farm rates in columns T–X, labelled "per day". Table 2a tests two readings of them against FOI.

Reading	Annual 2025 (kg)	÷ FOI	Implied t/ha
Col. U as kg per month × 6 months (Dec–May)	1,099,260	1.03	41.8
Col. V (= U × 30) as monthly × 6 months	32,977,800	30.8	1,254
FOI 2025 (reference)	1,069,740	1.00	40.7

Read as kg per month, column U gives 183,210 kg a month and 1,099,260 kg over six months, within 2.8% of FOI. The season length that would match FOI exactly is 5.84 months. The implied yield is 41.8 t/ha, and the primary-based x is 148.35 kg/bed against 144.36 from FOI, so two independent sources agree. Column V multiplies U by 30 and is 31 times FOI, so it is the inconsistent column. The per-bed value of 24.7 kg also fits a monthly reading: a month's worth is about 0.7 kg/m² on a 35.5 m² bed. The sheet's "per day" label cannot be checked from the file, so the unit should be confirmed with the source. Against the 2021 provincial-government total for Alaminos (5,208 MT), the primary figure is 21%, in line with FOI's 20.5%. Because the rates carry no year they are applied to 2025 only.

6. Results

6.1 Annual baseline and estimated salt beds (2025)

Table 3. Annual production and estimated salt beds

Municipality	Annual source	2025 output (MT)	Estimated beds	Range ($x_{2026} - x_{2025}$)	Beds at legacy x	Confidence
Alaminos City	Primary	1,099.3	7,410 (observed)	7,410	3,444	High (beds observed; output primary)
Anda	FOI	1,332.2	9,228	7,435 – 9,228	4,289	Low–Medium
Bani	FOI	607.8	4,210	3,392 – 4,210	1,957	Low–Medium
Bolinao	FOI	4,461.9	30,907	24,903 – 30,907	14,365	Low–Medium
Dasol	FOI	776.4	5,378	4,333 – 5,378	2,500	Low
Infanta	FOI	1,436.6	9,951	8,018 – 9,951	4,625	Low–Medium
San Fabian	Primary	106.0	734 (bed-eq.)	592 – 734	341	Annual High; beds Low
Total		**9,820.2**	**67,819**	**56,084 – 67,819**	**31,521**	

Beds at legacy x use 310.61 kg/bed/yr (3,444-bed basis) and are shown only to reconcile with the handwritten cards. Alaminos legacy value is the 3,444 count itself.

Bolinao contributes 45.4% and Infanta 14.6% of the 9,820.2 MT total. The total is 0.7% above the FOI provincial total (9,755 MT): San Fabian's and Alaminos' primary values replace their FOI values. Setting Assumptions!B11 = 2 (Bani primary) would add about 7,552 MT. The 2024 set (2,273.9 MT, Alaminos on FOI) is in the workbook with a low-confidence flag for FOI-based municipalities; Alaminos 2024 is FOI-only and understates, as the 2024 FOI series does everywhere (T6).

Monthly estimates. Table 4 (next page) gives 2025 in metric tons; the workbook holds 2024 and 2025 in kilograms, as a matrix (Monthly_Estimates) and in long format (Monthly_Long) ready to import into the PangAsin database.

6.2 Estimated monthly production by municipality, 2025 (MT)

Table 4. Monthly salt production estimate, 2025 (method 1: observed 2025 provincial profile for every municipality)

Municipality	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Alaminos City	2.7	6.8	58.8	410.1	312.8	156.3	43.8	10.3	18.1	20.9	25.3	33.4	1,099.3
Anda	3.3	8.3	71.2	497.0	379.1	189.4	53.1	12.5	21.9	25.3	30.6	40.4	1,332.2
Bani	1.5	3.8	32.5	226.8	173.0	86.4	24.2	5.7	10.0	11.6	14.0	18.5	607.8
Bolinao	10.9	27.6	238.6	1,664.6	1,269.8	634.2	177.8	41.9	73.4	84.9	102.6	135.5	4,461.9
Dasol	1.9	4.8	41.5	289.7	221.0	110.4	30.9	7.3	12.8	14.8	17.9	23.6	776.4
Infanta	3.5	8.9	76.8	536.0	408.8	204.2	57.3	13.5	23.6	27.3	33.0	43.6	1,436.6
San Fabian	0.3	0.7	5.7	39.5	30.2	15.1	4.2	1.0	1.7	2.0	2.4	3.2	106.0
Total	**24.0**	**60.8**	**525.1**	**3,663.7**	**2,794.6**	**1,395.9**	**391.4**	**92.3**	**161.4**	**186.9**	**225.9**	**298.2**	**9,820.2**

Every municipality follows the observed 2025 provincial profile (April 37.3%, May 28.5%, June 14.2%), so no month is empty. San Fabian's recorded Nov–May season, Alaminos' Dec–May and Bani's Jan–Apr are workbook switches that would zero the other months.

Table 4a. San Fabian monthly estimate from the hybrid annual series (MT)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Basis
2023	4.1	1.0	9.2	26.0	24.4	10.5	2.2	0.9	1.2	1.3	1.4	1.7	84.0	Primary
2024	9.8	1.9	17.2	25.5	30.8	11.2	1.3	1.3	1.3	1.1	1.1	1.1	103.5	Primary
2025	0.3	0.7	5.7	39.5	30.2	15.1	4.2	1.0	1.7	2.0	2.4	3.2	106.0	Primary
2026	4.5	1.1	10.1	28.5	26.8	11.5	2.4	1.0	1.3	1.4	1.5	1.9	92.0	FOI fill

2024 and 2025 use the observed profile of the same year; 2023 and 2026 use the pooled 2024–25 profile because no full provincial monthly series exists for them. 2026 is provisional (FOI fill).

Peak month: April, 3,663.7 MT (37.3% of the year), followed by May. Because all municipalities follow the same profile, the estimated monthly pattern tracks the FOI provincial series; differences come only from the San Fabian and Alaminos primary values.

7. Why these are the most reliable assumptions available

Reliability is judged by whether a choice is traceable to primary evidence, passes an independent numeric check, and stays honest about what it cannot see. Table 5 sets each choice against the alternatives considered.

Table 5. Chosen assumptions versus alternatives

Decision	Chosen	Alternative considered	Evidence for the choice
Annual production	Alaminos and San Fabian: primary (San Fabian 2026 filled by FOI); FOI for the rest	FOI for Alaminos; primary for Bani; OPAg 2021 totals	Alaminos primary agrees with FOI within 3% (T10). Bani follows FOI because its primary may be an area norm (40 t/ha × area). OPAg is a different year and source class, so it only benchmarks (T9).
Bed-yield x	144.36 (Alaminos 2025, 7,410 beds)	310.61 (3,444 beds); master-sheet col. V; 2024; mean of 2025–26	Only full-census calibration (T2); matches 40–45 t/ha elsewhere (T3), and a primary-based calibration (148.35, T10); 2024 coverage is incomplete (T6); 2026 is provisional.
Calibration year	2025	2024, 2026	Only FOI year noted as assisted + non-assisted; province total reconciles to the monthly file within 2.8%.
Monthly shape	Observed profile for every municipality	Linear arithmetic (M1 = 8%); uniform	Error 23% versus 53% (T7); recorded seasons stay as switches (T8) because they would leave months empty.
Within a window	Flat split	Provincial shape renormalised	No monthly evidence for another shape; the provincial shape would put about 86% of Bani output in April (relevant only on its primary basis).
San Fabian beds	Bed-equivalent, area count shown	Literal area count only	Cooked salt is not area-bound (T5); both numbers are provided.

None of these choices is proven correct. They are the ones that rest on primary evidence, survive the tests above, and fail visibly rather than silently when the data change, because each is a selector or input in the workbook.

8. Limitations and threats to validity

- **FOI is a lower bound.** Coverage ranges from 3% (Dasol) to 37% (Bolinao) of the 2021 provincial figures; no municipal correction is applied (multiplier = 1.00).
- **Alaminos output rests on a unit reading.** Master-sheet column U is labelled per day but is used as kg per month, because that reading matches FOI within 2.8% and gives 41.8 t/ha (T10); the label should be confirmed. The six-month season is assumed (FOI matches at 5.84 months), and the rates carry no year, so they apply to 2025 only. Alaminos 2024 is FOI-only and not comparable.
- **One calibration point.** x rests on Alaminos 2025 alone. The 144–179 band reflects two years, not a statistical interval.
- **Bed size is not uniform.** Dasol banigan average about 2,400 kg per unit against 144 kg for an Alaminos bed (T9). Bed counts outside Alaminos are Alaminos-size bed-equivalents until local surveys exist.

- **Bani's source is unsettled.** FOI gives 608 MT and the master sheet 8,160 MT (13×), and the 2025 master-sheet values equal exactly 40 t/ha × area. FOI is used by decision; if the master sheet is right, Bani is under-estimated by about 7,550 MT and about 52,000 beds.
- **Monthly shapes are provincial.** The only monthly data are provincial aggregates for two years and are Bolinao-dominated; municipal seasons may differ.
- **2026 is provisional.** The FOI cut-off date is not stated, so 2026 is excluded from the monthly output and used only for sensitivity.
- **Inconsistent FOI notes.** The 2024 monthly file (noted "assisted only") totals 3.9% more than the 2024 municipal file (noted "assisted and non-assisted"); 2025 differs by 2.8%.
- **The data are synthetic.** Monthly values are deterministic disaggregations of annual totals and add no information. They suit pipeline development, demonstrations and sensitivity analysis. A forecasting model trained on them would only learn the assumed profile, so they must not be reported as observations.

9. Conclusions and recommendations

Conclusions

1. The 2025 annual baseline for the seven municipalities is 9,820.2 MT, of which Alaminos City contributes 1,099 MT from primary master-sheet rates, 2.8% above its FOI value (T10). San Fabian is primary (hybrid series for 2023–2026, FOI filling 2026); Anda, Bani, Bolinao, Dasol and Infanta are FOI-based lower bounds.
2. The bed-yield constant is 144.36 kg per bed per year. The handwritten 310.61 used a partial denominator and should be replaced; the corrected value also agrees with independent yields of 40–45 t/ha.
3. Estimated salt beds total about 67,800, of which 7,410 (Alaminos City) are observed. Bolinao, Infanta and Anda carry the largest estimated counts (Bani would rise to about 56,500 on its primary basis); San Fabian's count is a bed-equivalent only.
4. Linear arithmetic with M1 = 8% is not supported: it performs no better than a uniform split and ignores recorded production seasons. An observed seasonal profile with primary windows is the best-supported monthly method now available.
5. Confidence is highest for Alaminos beds and San Fabian annual output, low-to-medium for Bani (source disputed) and the other FOI-based totals, and low for Dasol.

Recommended next data

- Obtain the Alaminos 2024 bed count and farm-level harvest to settle the 2024 → 2025 question (T6).
- Ask the Alaminos agriculture office to confirm that master-sheet column U is kg per month and how many months a year production runs.
- Request OPAg municipal series for 2022–2025 and replace FOI baselines where they are more complete.
- Measure bed dimensions on about ten farms each in Dasol, Bolinao, Anda and Infanta to validate or replace x per municipality.
- Collect monthly harvest logs from at least two farms per municipality through the ASIN Center to replace the provincial profile.

- Confirm with Bani's municipal office whether the 2025 reports are measured harvests, to decide whether the primary basis should replace FOI.
- Record San Fabian output as cooked batches (pans × batches) instead of beds, and ask BFAR RFO1 which note on the 2024 files is correct.

Appendix A. Reconciliation of the handwritten cards

Card	Handwritten	Recomputed	Comment
1 · x, Alaminos 2025	1,069,740 ÷ 3,444 = 310.61	÷ 7,410 = 144.36	Arithmetic correct; denominator is a partial count (T2)
2 · Daily rate	1.18 kg/bed/day	310.61 ÷ 365 = 0.85	Reproduced only with 12 × 22 working days (1.18); not from calendar days
3 · Anda 2025 beds	1,332,211 ÷ 310.61 = 4,289.02	4,289.04 unrounded; 9,228 at x = 144.36	Rounding only
4 · Alaminos 2024	446,350 ÷ 310.61 = 1,437; mean 2,440.5	446,350 ÷ 144.36 = 3,092; mean 5,251	Card prints 360.61, read as 310.61
5 · San Fabian 2024	96,000 → 309.07	665.0 at x = 144.36	FOI value; primary is 103,500
5 · San Fabian 2025	70,495 → 226.96; "new" 104,000 → 341.26	106,000 → 734.3 at x = 144.36	341.26 = 106,000 ÷ 310.61, so the email's 106 MT was used, not 104,000
5 · San Fabian 2026	91,975 → 296.11	637.1 at x = 144.36	Provisional FOI value
5 · Means	277.38 (FOI only); 315.48 (new)	(665.0+734.3+637.1)/3 = 678.8	Card labels these "no. of salt beds"; they are mean bed estimates
6 · January share	8% → 106,568 kg	0.08 × 1,332,211 = 106,577 kg	Difference 9 kg (0.008%), rounding

Appendix B. Using the workbook

- **Assumptions!B4** selects the bed-yield basis: 1 = Alaminos 2025 (default), 2 = mean of 2025 and 2026, 3 = legacy 310.61.
- **Assumptions!B5** selects the monthly method: 1 = same-year observed profile with windows (default), 2 = pooled 2024–25 profile with windows, 3 = linear arithmetic.
- **Assumptions!B23** chooses Alaminos' annual source (2 = master col. U read as kg per month × 6 months, default; 1 = FOI). **C13** and **C19** switch on the recorded Alaminos (Dec–May) and San Fabian (Nov–May) seasons; **B4 = 4** calibrates x on the Alaminos primary production. **Assumptions!B24** sets how San Fabian's missing years are filled (1 = FOI as reported, 2 = FOI scaled by the overlap ratio).
- **Assumptions!B11** chooses Bani's annual source (1 = FOI, default; 2 = primary master sheet and Jan–Apr window).
- **Assumptions!B10** sets the shape inside a primary window (1 = equal months, 2 = observed provincial profile). **Monthly_Override** accepts observed monthly kg for any municipality-year and replaces the estimate for that row.
- **Assumptions!B6** sets a_1 for the linear method; the municipality table holds the coverage multipliers and season-window flags.
- Blue cells are inputs from source files, black cells are formulas, green cells link to other sheets. The Testing sheet recomputes T1–T10 whenever an input changes.

Appendix C. References

[1] Pangasinan Provincial Government. "Dasol is still Pangasinan's top salt producer" (OPAg 2021 municipal production, salt-farm area and banigan counts). https://www.pangasinan.gov.ph/?p=108708

[2] Philippine Information Agency, Pangasinan. “PangASINan, a trailblazer in salt production,” 31 May 2023. https://mirror.pia.gov.ph/features/2023/05/31/pangasinan-a-trailblazer-in-salt-production

[3] Philippine Daily Inquirer. “COVID brings Pangasinan town’s salt industry to a standstill,” 28 September 2020 (harvest period December–May). https://newsinfo.inquirer.net/1340841/covid-brings-pangasinan-towns-salt-industry-to-a-standstill

[4] BFAR Regional Fisheries Office No. 1, Development of Salt Industry Project. FOI releases RE-RS3969-PSU and RE-RS4760-ATTACHMENT (production 2023–2026; active producers).

[5] salt_MasterSheet1 (Bani and Alaminos City survey data); San Fabian email profile; handwritten calculation cards (Linear-Arithmetic-to-assume-monthly-production).

Appendix: Stronger IQR Adjustment to 2024-2025 Monthly Estimates

This is a correction to estimated monthly allocations for 2024 and 2025. It does not change any annual production totals and does not create a 2026 prediction.

Alaminos-calibrated stochastic IQR method

For each calendar month, calculate Alaminos City log change g = ln(2025 production / 2024 production) using the same month. Calculate Q1, median, and Q3 across the 12 values. For every non-Alaminos municipality-year, independently draw U_m uniformly from [Q1,Q3] and calculate a positive noise multiplier r_m = exp(k ? (U_m ? median)), where k = 3.0 amplifies IQR variability. Reweight the original monthly value A_m and normalize: adjusted_m = A_m ? r_m ? T / sum_j(A_j ? r_j), where T is that municipality-year's original annual total. Consequently, sum_m(adjusted_m) = T exactly (apart from floating-point rounding).

Calibration: Q1=0.4332, median=1.1762, Q3=1.5227; at k=3, pre-normalization multipliers range from 0.108x to 2.828x. Seed=42. Alaminos City remains unchanged; the other six municipalities receive independent month-specific adjustments in both years.

Annual totals and monthly-share variability

Municipality	Year	Annual total (kg)	Original share CV	Adjusted share CV
Alaminos City	2024	446,350	1.177	1.177
Alaminos City	2025	1,099,260	1.402	1.402
Anda	2024	115,600	1.177	1.263
Anda	2025	1,332,211	1.402	1.103
Bani	2024	649,400	1.177	1.096
Bani	2025	607,800	1.402	1.532
Bolinao	2024	21,825	1.177	1.865
Bolinao	2025	4,461,917	1.402	1.827
Dasol	2024	579,267	1.177	1.180
Dasol	2025	776,414	1.402	1.382
Infanta	2024	357,916	1.177	1.803
Infanta	2025	1,436,623	1.402	0.960
San Fabian	2024	103,500	1.177	1.068
San Fabian	2025	106,000	1.402	2.225

Evidence limitation: the stochastic spread is calibrated from Alaminos City, not validated against observed monthly production in the other municipalities. The 3x amplitude is a deliberate variation setting, not a claim that the resulting values are measured or proven close to actual monthly output. Municipality-specific observed monthly data would be needed to assess that.

Adjusted Monthly Historical Estimates (kg)

Alaminos City remains unchanged as the reference. Other municipalities show stronger stochastic IQR variation; all annual totals remain fixed.

2024

Municipality	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual total
Alaminos City	42,118	8,036	74,349	109,914	132,793	48,460	5,421	5,488	5,545	4,796	4,748	4,682	446,350
Anda	13,573	866	31,594	27,576	4,639	30,188	1,675	1,840	216	537	410	2,486	115,600
Bani	178,940	5,066	114,054	42,322	73,371	150,762	20,635	43,259	5,367	5,371	7,352	2,902	649,400
Bolinao	2,834	92	1,033	814	12,568	3,077	393	509	180	222	54	49	21,825
Dasol	190,486	4,799	25,006	76,643	96,367	117,209	9,307	19,775	13,516	5,032	18,881	2,246	579,267
Infanta	7,975	621	6,345	147,952	150,162	27,687	750	7,565	4,102	2,898	1,197	662	357,916
San Fabian	20,386	4,822	7,029	30,693	10,914	17,472	1,143	631	558	2,348	595	6,908	103,500

2025

Municipality	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual total
Alaminos City	2,691	6,811	58,779	410,106	312,825	156,256	43,814	10,331	18,072	20,916	25,282	33,377	1,099,260
Anda	5,267	23,919	59,738	205,623	457,232	45,924	156,324	19,427	51,371	15,898	143,981	147,509	1,332,211
Bani	287	2,252	8,617	255,704	91,177	165,905	30,177	2,003	19,162	20,271	6,264	5,981	607,800
Bolinao	11,286	14,988	175,933	2,358,218	1,175,057	450,214	128,582	13,164	9,431	41,132	24,054	59,859	4,461,917
Dasol	503	1,585	108,064	321,819	91,900	139,364	12,500	17,444	13,471	12,601	11,746	45,415	776,414
Infanta	1,285	45,346	90,837	276,838	296,264	361,496	27,203	59,181	75,126	76,527	36,160	90,359	1,436,623
San Fabian	184	217	4,153	4,525	73,506	7,741	5,795	139	908	1,065	5,551	2,215	106,000

2026 Production: Jan-Sep San Fabian Reference Synthesis

Reporting period correction. The values in the provided 2026 production column represent cumulative production through September (January-September), not full-year annual production. The displayed province total, 14,981,789 kg, is the supplied Jan-Sep total.

Jan-Sep synthesis method

The workbook contains a synthetic San Fabian 2026 Jan-Dec profile. Its January-September pattern is normalized to a nine-month profile and used as the reference. Other municipalities use that pattern with independent stochastic IQR perturbations. Each municipality profile is normalized across Jan-Sep and multiplied by its supplied Jan-Sep total.

z_m = ln(reference share_m / median reference share). For the nine San Fabian Jan-Sep shares: Q1=-1.210220, median=0.000000, Q3=0.951565. Other-municipality multiplier: exp(3*(U_m-median(z))), where U_m is independently drawn from Uniform(Q1,Q3); seed=42.

Reconciliation: The nine generated monthly values for each municipality sum to that municipality's supplied Jan-Sep amount. October-December are excluded. No full-year annualization is implied.

Interpretation: The supplied Jan-Sep aggregate amounts come from the provided image. The monthly municipality-level breakdown is synthetic, based on the San Fabian profile and IQR variation; it is not observed month-by-month production.

Jan-Sep 2026 Monthly Synthesis (kg)

Synthetic monthly allocations using the San Fabian reference profile and stochastic IQR variation. The Jan-Sep row totals reconcile to the supplied cumulative production amounts through September; October-December are not included.

Municipality	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Jan-Sep total
Alaminos City	68,974	1,963	271,434	268,514	5,048	660,613	34,095	16,718	312	1,327,670
Anda	10,501	1,567	524,291	235,605	706,521	25,994	1,327	4,625	255	1,510,686
Bani	3,281	230	4,750	976	49,885	12,992	1,280	12	94	73,500
Bolinao	3,281	1,679	471,058	1,978,370	7,884,192	52,919	14,661	11,630	2,516	10,420,307
Dasol	1,121	2,639	4,779	237,800	49,428	276,816	24,327	820	31,781	629,511
Infanta	208,756	3,485	16,659	604,308	16,814	10,698	637	41,666	25,117	928,140
San Fabian	4,697	1,174	10,676	30,048	28,241	12,164	2,523	1,052	1,400	91,975
TOTAL	**300,611**	**12,737**	**1,303,647**	**3,355,621**	**8,740,129**	**1,052,196**	**78,850**	**76,523**	**61,475**	**14,981,789**

The aggregate amounts are from the provided 2026 table. Monthly values are generated estimates, not observed monthly records.

Appendix: 2026 Jan-Sep Forecast Comparison

Holdout design: each municipality model was trained on 24 monthly values (January 2024-December 2025) from the IQR-adjusted workbook. Predictions cover January-September 2026 and are compared with the generated San Fabian-based monthly synthesis. The aggregate comparison values are the supplied Jan-Sep totals; the monthly allocations are synthetic.

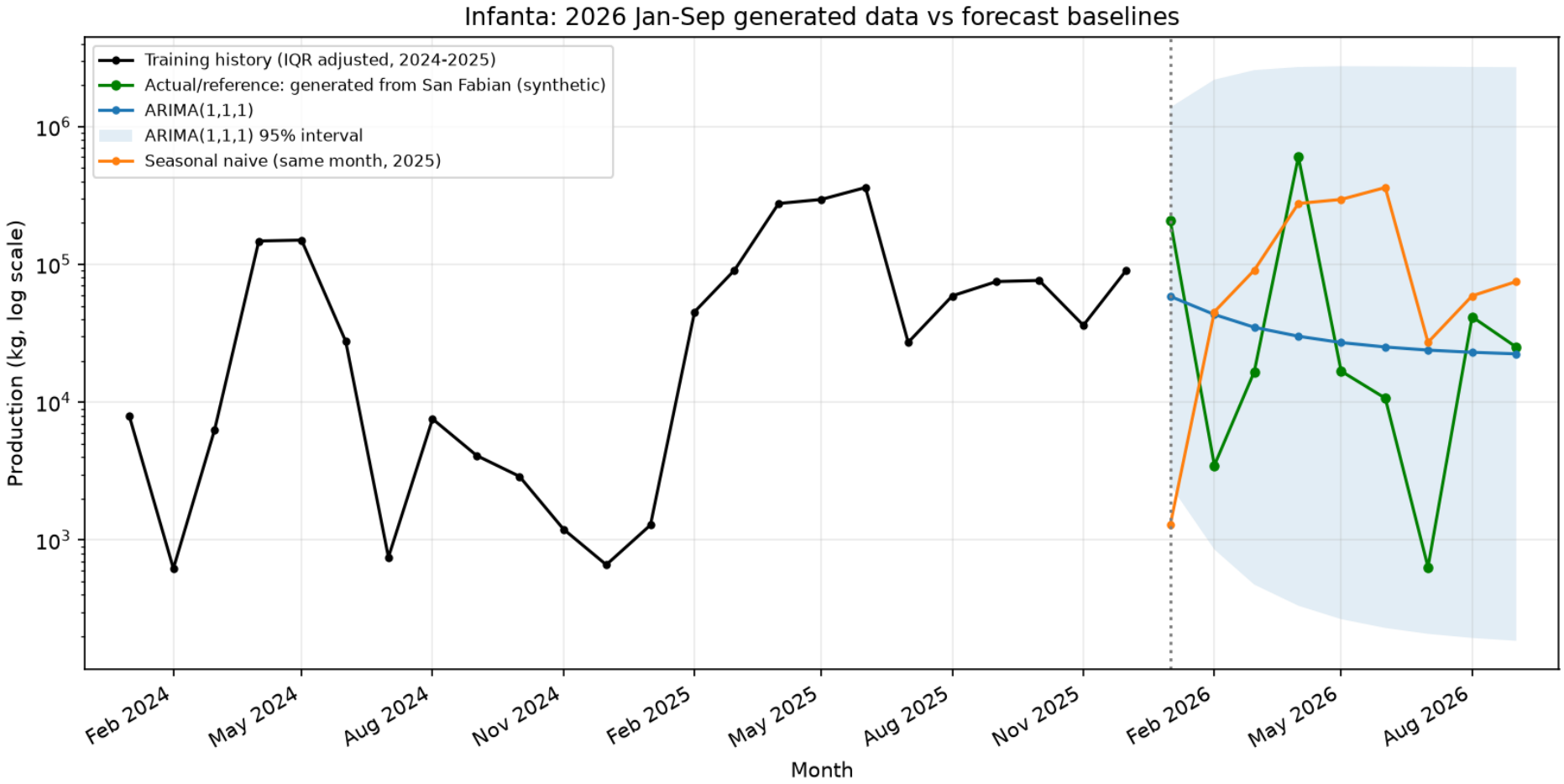
Municipality	Model	MAE (kg)	RMSE (kg)	WAPE	Bias (kg)
Alaminos City	ARIMA(1,1,1)	138,597	238,326	94.0%	-114,817
Alaminos City	Seasonal naive (same month, 2025)	141,264	215,830	95.8%	-34,220
Anda	ARIMA(1,1,1)	215,793	251,630	128.6%	7,259
Anda	Seasonal naive (same month, 2025)	112,473	184,582	67.0%	-53,985
Bani	ARIMA(1,1,1)	15,294	17,032	187.3%	8,105
Bani	Seasonal naive (same month, 2025)	56,419	100,667	690.8%	55,754
Bolinao	ARIMA(1,1,1)	1,158,533	2,689,115	100.1%	-1,097,847
Bolinao	Seasonal naive (same month, 2025)	880,565	2,246,355	76.1%	-675,937
Dasol	ARIMA(1,1,1)	66,737	111,775	95.4%	-44,005
Dasol	Seasonal naive (same month, 2025)	46,185	65,977	66.0%	8,571
Infanta	ARIMA(1,1,1)	94,638	198,706	91.8%	-71,047
Infanta	Seasonal naive (same month, 2025)	152,813	200,611	148.2%	33,937
San Fabian	ARIMA(1,1,1)	8,375	12,990	81.9%	-7,135
San Fabian	Seasonal naive (same month, 2025)	10,209	17,624	99.9%	577
ALL MUNICIPALITIES	ARIMA(1,1,1)	242,567	1,028,445	102.0%	-188,498
ALL MUNICIPALITIES	Seasonal naive (same month, 2025)	199,990	860,382	84.1%	-95,043

Pooled across 63 municipality-month comparisons, the seasonal-naive baseline had WAPE 84.10% and MAE 199,990 kg; ARIMA(1,1,1) had WAPE 102.00% and MAE 242,567 kg. The baseline performed better overall, but not for every municipality. For Infanta, ARIMA MAE was 94,638 kg versus 152,813 kg for the baseline.

A previous SARIMA(0,1,1)(0,1,1,12) experiment was not retained in the final comparison: two years of training provide only two annual cycles, and the model produced unstable, extremely large forecasts for some municipalities, including Infanta. Do not interpret the synthetic holdout scores as measured real-world accuracy.

Infanta: Generated Jan-Sep Series and Forecasts

The green line is the Jan-Sep monthly allocation generated using the San Fabian reference profile and stochastic IQR method. Blue is ARIMA(1,1,1); orange is the seasonal-naive baseline (each month repeats the corresponding 2025 value). The shaded band is the ARIMA 95% interval; the baseline has no fitted interval.



Interpretation and limitations. The monthly green values are generated estimates, not independent observations. The IQR noise was intentionally introduced to vary monthly allocations while preserving the supplied Jan-Sep total. Consequently, monthly swings can be unpredictable from the historical training series. More observed monthly records (and, ideally, more than two years) are needed to assess real predictive accuracy. WAPE/MAE are more useful than MAPE here because some monthly reference values are small.